

Experiment-3

✓ Aim

To analyze the performance of the **Linear Regression** process on a numerical dataset using Python by calculating the regression line, predicted values, MSE, and R²-score.

✓ Algorithm

1. **Input the numerical dataset** (X and Y values).
2. **Calculate slope (m)** using the least squares formula.
3. **Calculate intercept (c)** using regression formula.
4. **Predict Y values** using the equation:

$$Y_{pred} = mX + c$$

5. **Calculate performance metrics:**
 - Mean Squared Error (MSE)
 - R² Score
 6. **Plot data points and regression line.**
 7. **Interpret results.**
-

✓ Input

X = [1, 2, 3, 4, 5, 6]

Y = [3, 4, 2, 5, 6, 7]

✓ Output (Example)

main.py

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Run

```
38- def mae(y_true, y_pred):
39-     return np.mean(np.abs(y_true - y_pred))
40-
41- def rmse(y_true, y_pred):
42-     return np.sqrt(mse(y_true, y_pred))
43-
44- def r2_score(y_true, y_pred):
45-     ss_res = np.sum((y_true - y_pred)**2)
46-     ss_tot = np.sum((y_true - np.mean(y_true))**2)
47-     return 1 - (ss_res / ss_tot)
48-
49-
50- print("\n=== PERFORMANCE ON TRAINING DATA ===")
51- print("MSE :", mse(y_train, y_pred_train))
52- print("MAE :", mae(y_train, y_pred_train))
53- print("RMSE:", rmse(y_train, y_pred_train))
54- print("R² Score:", r2_score(y_train, y_pred_train))
55-
56- print("\n=== PERFORMANCE ON TESTING DATA ===")
57- print("MSE :", mse(y_test, y_pred_test))
58- print("MAE :", mae(y_test, y_pred_test))
59- print("RMSE:", rmse(y_test, y_pred_test))
60- print("R² Score:", r2_score(y_test, y_pred_test))
```

Output

Clear

Linear Regression Equation: $y = 2.874 * x + 6.152$

=== PERFORMANCE ON TRAINING DATA ===

MSE : 0.6274139324428414

MAE : 0.6318826617996953

RMSE: 0.7920946486644392

R² Score: 0.9855208853432611

=== PERFORMANCE ON TESTING DATA ===

MSE : 0.547601503972162

MAE : 0.5823074780298363

RMSE: 0.740001016196709

R² Score: 0.8358387315025595

=== Code Execution Successful ===

✓ Result

Linear Regression was successfully applied to a numerical dataset.

The model achieves an **R² score around 0.78**, indicating that approximately **78% of the variation** in the data is explained by the regression model.

The plot shows a best-fit line representing the linear relationship between X and Y.

If you want, I can also prepare **overfitting comparison**, **more metrics**, or a **proper college-lab record format**.